ISHAN JOSHI

AI ENGINEER | GENERATIVE AI • NLP • MACHINE LEARNING • MLOps

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PROFESSIONAL SUMMARY

AI Engineer with an M.Tech in Data Science and hands-on experience across machine learning, NLP, computer vision, multimodal AI, and GPU infrastructure. Build and deploy AI/LLM applications using Python, PyTorch, TensorFlow, Docker, Kubernetes, and Ollama, including a multi-agent clinical decision-support platform on NVIDIA DGX A100. Experienced in extracting insights from structured and unstructured data, developing analytical workflows, and translating model outputs into interactive applications and business-facing demonstrations.

TECHNICAL SKILLS

Generative AI & NLP: LLM applications, multi-agent AI, Ollama, BERT, MiniLM, sentence embeddings, NLP, text classification, information extraction

Machine Learning: Scikit-learn, classification, clustering, predictive modeling, feature engineering, hyperparameter tuning, model evaluation

Deep Learning: PyTorch, TensorFlow, CNNs, VGG-16, ResNet-50, transfer learning, multimodal learning, cross-modal attention

Data & Analytics: Python, SQL, Power BI, MS Excel, interactive visualization, structured/unstructured data analysis

MLOps & Infrastructure: Docker, Kubernetes, NVIDIA DGX A100, MIG, GPU utilization, pod management, Grafana, job scheduling, Streamlit

Programming: Python, Java, SQL, HTML/CSS, JavaScript

PROFESSIONAL EXPERIENCE

AI Engineer | Amity University, Noida | *June 2025 – Present*

- Manage containerized AI research workloads across two NVIDIA DGX A100 servers using Kubernetes and Docker, including pod lifecycle, GPU allocation, job scheduling, and resource monitoring for 30+ concurrent AI workloads.
- Designed Kubernetes pod monitoring and automated fault-recovery workflows with Grafana, reducing unresponsive-pod incidents and improving platform uptime.
- Engineered and hosted a multi-agent clinical decision-support platform using Ollama and Docker on an NVIDIA DGX A100 MIG 7g.40gb partition, maintaining 99%+ uptime while optimizing VRAM utilization.
- Develop and optimize AI models with Python, TensorFlow, and PyTorch using feature engineering, hyperparameter tuning, and model evaluation.
- Deploy secure Streamlit AI/LLM applications for real-time inference and interactive data visualization used in industry demonstrations and research workflows.
- Collaborate with faculty and research teams to translate analytical and AI requirements into deployable solutions and automated workflows.

Product Pricing Analyst | PlanCover, Gurgaon | *Apr 2023 – Aug 2023*

- Performed market and competitor analysis for health-insurance pricing using Excel and Power BI, converting claims and market data into actionable pricing insights.
- Applied Impact Factor metrics to claims MIS data for customer segmentation and identification of high-risk policyholders, supporting risk-adjusted pricing and underwriting decisions.
- Collaborated with the technology team on a Quote Management System that streamlined quote generation across multiple insurers and improved coordination between sales and underwriting.

Content Manager | Aapki Education | *Jan 2021 – Feb 2023*

- Managed STEM content planning and delivery while using learner engagement data to optimize content formats, difficulty, and sequencing.
- Analyzed completion rates, time-on-task, click-through rates, drop-off rates, and session duration to drive data-informed content decisions.
- Collaborated with technology teams to design performance dashboards supporting content lifecycle management.

Programmer Analyst Trainee | Cognizant Technology Solutions, Chennai | *Jul 2020 – Nov 2020*

- Developed responsive internal e-library interfaces using HTML, CSS, JavaScript, and Bootstrap within an agile team.
- Supported UI/UX improvements with attention to responsive design and accessibility.

KEY PROJECTS

LLM-Driven Workflow for Competitive Insight Extraction

- Built an NLP analytics pipeline using MiniLM sentence embeddings and K-Means clustering to extract and organize insights from KPMG and PwC research reports.
- Mapped clustered content to Finance, Technology, and ESG domains for competitive benchmarking and surfaced results through interactive sunburst and similarity visualizations.

Deep Learning Facial Emotion Recognition

- Implemented VGG-16 and ResNet-50 models on FER-2013 for facial-emotion classification, applying normalization, resizing, augmentation, and comparative model evaluation.
- Assessed accuracy versus computational cost to identify a model suitable for real-time inference.

Restaurant Review Sentiment Analysis

- Built an NLP classification workflow for positive, neutral, and negative restaurant reviews using tokenization, lemmatization, Naive Bayes, and SVM.
- Compared classifier performance to identify an effective text-classification approach.

EDUCATION

M.Tech in Data Science | Amity University, Noida | Aug 2023 – Jun 2025 | CGPA: 8.98

B.Tech in Civil Engineering | Vellore Institute of Technology, Chennai | Jul 2016 – Jun 2020 | CGPA: 8.36

PUBLICATIONS

Emotion Recognition Using BERT and CNN: A Deep Learning Approach for Crossmodal Fusion of Text and Audio — ICCCNet 2025, Manchester Metropolitan University. DOI: 10.1007/978-3-032-18910-3_19. Developed a multimodal model combining BERT text representations and CNN-based mel-spectrogram features with cross-modal attention and fusion.

Predictive Modelling of Sleep Efficiency Using Machine Learning — ICETM 2025. DOI: 10.1109/ICETM63734.2025.11051741. Developed Random Forest, Gradient Boosting, and related models; achieved $R^2 = 0.85$ and MSE ≈ 0.0027.

Hybrid Approach to Digital Image Watermarking — CRM2025, SR University. DOI: 10.1007/978-981-96-9771-7_28. Developed a hybrid multi-domain approach for robust watermarking with improved security and imperceptibility.

CERTIFICATIONS

- IIT Delhi — AI with Machine Learning
- NVIDIA — Deep Learning Certificate Program
- Udemy — Power BI Masterclass
- Udemy — Machine Learning A-Z: AI, Python & R